Christos **Hadjinikolis**

Machine Learning Engineering Manager | Production ML & Applied AI Systems
Ph.D. Computer Science | UCL Research Associate | AI Standards Expert @ ISO/CEN-CENELEC JTC'21
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profiles

website
LinkedIn
GitHub
Google Scholar

leadership

6 direct reports
performance/career
delivery ownership
hiring/mentoring

applied AI

LLM/tool use
model eval/replay
streaming UX
approval gates
AI governance

AI stack

PyTorch
Hugging Face
MLflow
SentencePiece
faster-whisper/Kokoro

engineering

model-serving
Python/Java
Kotlin/SQL
Kafka/Flink
reliability

open source

PyPI: dynamicio
skeleton-replay
JetBrains plugin

about me

ML Engineering Manager with **16 years** across applied ML systems, data platforms, backend services, and AI research. I lead cross-functional teams that translate research-grade models and ambiguous product requirements into reliable, high-throughput production systems. Recent work spans **model-serving**, offline/online evaluation, streaming ML infrastructure, and reliability for high-volume prediction pipelines, backed by a Ph.D. in Computer Science and active AI standards work with ISO/CEN-CENELEC.

work-experience

Vortexa Ltd, London, UK 12/2020–present — Engineering Manager | ML Systems Lead

Own engineering strategy and delivery for a live ML/data estate turning **6M vessel-position records/hour** into production intelligence for **13.5K monitored vessels**. Manage **6 direct reports** and lead a **10-person** cross-functional team accountable for model quality, stakeholder alignment, reliability, and delivery maturity.

- **People & Delivery:** Manage **6 direct reports** spanning Product, SME analysis, Data Science, and Data Engineering; own performance/career development, team delivery, hiring for **6 roles**, and delivery accountability for the wider team.
- **Model Development & MLOps:** Led **0-to-1** research-to-production delivery for destination and ETA/waiting-time sequence/transformer models in **PyTorch**; established **MLflow**, model/data versioning, automated evaluation gates, replay, monitoring, and model-serving workflows at **6M events/hour**.
- **Evaluation & Iteration:** Built offline/online model-evaluation and replay loops; analysed edge cases and failure modes with domain experts and Product to drive metric-guided model, data, and interface improvements.
- **Product/Stakeholder Alignment:** Ran workshops across Product, SMEs, analysts, and engineers to turn model-quality disputes into shared definitions, measurable objectives, interface/SLA proposals, and OKR-linked workstreams.
- **Platform & Standardisation:** Standardised I/O seams, data contracts, repo archetypes, AWS CodeArtifact publishing, ADRs, dev containers, and local E2E tests; cut a **3h DAG feedback loop to under 5 min**.
- **Architecture & Reliability:** Led **Kafka Streams-to-Flink** as a strategic platform move; maintained signal-quality controls, rollback/fallback paths, shared on-call, and **MTTR under 30 min**.

DataReply, London, UK 04/2016–12/2020 — Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant and helped grow it from **three practitioners and two managers to 30+ consultants**. Progressed to Senior Consultant; shaped opportunities, agreed deliverables/timelines, managed expectations, placed consultants, and delivered technical roadmaps across UBS, Vodafone, CNHi, and other enterprise clients.

- **Vodafone Group:** Led multi-workstream GCP data-science platform delivery; drove requirements, migration planning, CI/CD, workflow automation, ML lifecycle support, and feature-engineering migration.
- **CNHi:** Lead Data Scientist/Scrum Master on predictive maintenance; translated stakeholder needs into DS/DE backlog and guided a **PySpark** team over live telemetry.
- **UBS:** Embedded consultant who grew from Data Scientist into ML Engineer through XP/pairing while delivering graph/process-mining analytics, observability, and real-time **Kafka**/Elasticsearch pipelines.

KCL | UCL | David Game College, UK 2010–2016 — Academic Teaching & Technical Instruction

Taught **algorithm design**, **data structures**, databases/**SQL**, and Java/Python during and after doctoral research; developed presentation discipline, learner support, and mentoring habits.

education

2010–2014 **Ph.D. in Computer Science** King's College London, UK
***Thesis**: Persuasion Dialogues & Opponent Modelling.*
Developed logical inference methods for large **Knowledge Graphs**, combining symbolic reasoning, **graph analytics**, and Bayesian techniques (e.g., MCMC). Received **Best Poster Award** at IJCAI 2013 (413 submissions), **EPSRC PhD Scholarship**, **Outstanding TA Award**, and **Graduate Certificate in Academic Practice**.
Access Publications on Google Scholar

2004–2010 **Diploma (BEng) in Computer Engineering** University of Thessaly, GR
Five-year polytechnic degree in Computer Science and Engineering with a **strong focus on mathematics** (10 courses), covering theoretical aspects, networking, and telecommunications. ***Majored in Artificial Intelligence***.

2002–2004 **Military Officer: Second Lieutenant** National Guard, CY
Completed national service as a commissioned Second Lieutenant.

selected AI/tooling work

Private **Promet** GenAI harness
Private applied GenAI project exploring voice, memory, tool-use, and streaming assistant workflows; hands-on work on local runtimes, faster-whisper STT, Kokoro/**Hugging Face** assets, tool routing, schema validation, approval gates, durable state, traces, and deterministic replay/evaluation for auditable LLM interactions.

PyPI **skeleton-replay** Runtime evidence
Python developer tool that turns script/pytest runs into traces, architecture snapshots, workflow evidence, and replayable reports for review, debugging, onboarding, and LLM-assisted code understanding.

JetBrains **Skeleton Replay plugin** IDE workflow
Free PyCharm/IntelliJ plugin bringing Skeleton reports, workflow evidence, and source navigation into the IDE; reflects a bias for developer experience, evidence, and shared understanding.

PyPI **dynamicio** Data contracts
Free public tooling for making I/O seams explicit, validating schema/data expectations, and switching local/dev/prod datasets so pipeline logic can be tested with faster feedback.

committees/affiliations

since 10/2024 **University College London, Dept. of Information Studies** Associate Researcher
Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.

since 01/2021 **ISO - International Organisation for Standardisation** Committee Expert Member for JTC'21
Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.

talks/interviews

2023 **Bill Raymonds, Agile in Action** Podcast Interview
Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa, **Details:** YouTube

2022 **Open Data Science Conference** Industry Conference Talk
dynamicio, a published **PyPI library** for abstracting I/O in ML/data workflows, **Details:** here

2020 **iunera: Big Data Innovators** Interview Blog Post
The Agile approach in data science explained by an ML expert, **Details:** here

2020 **Big Data Warsaw** Industry Conference Talk
Monitoring & Analysing Communication and Trade Events as Graphs, **Details:** here

2018 **Connected Data London** Panelist @ Industry Conference
AI tribes and interconnections: what's graph got to do with it?, **Details:** here

2018 **Minds Mastering Machines** Industry Conference Talk
A talk on doing Data Science the Agile Way, **Details:** here

certificates & awards

2017–2020 **Selected ML/cloud credentials** AWS, Google Cloud, O'Reilly
AWS ML Specialty; Google Professional Data Engineer; Apache Spark Developer.

2016–2019 **Data/graph engineering credentials** Coursera, Elastic, Neo4j
Process Mining; Graph Analytics; Advanced Elasticsearch; Neo4j Certified Professional.